

BODY TEMPERATURE

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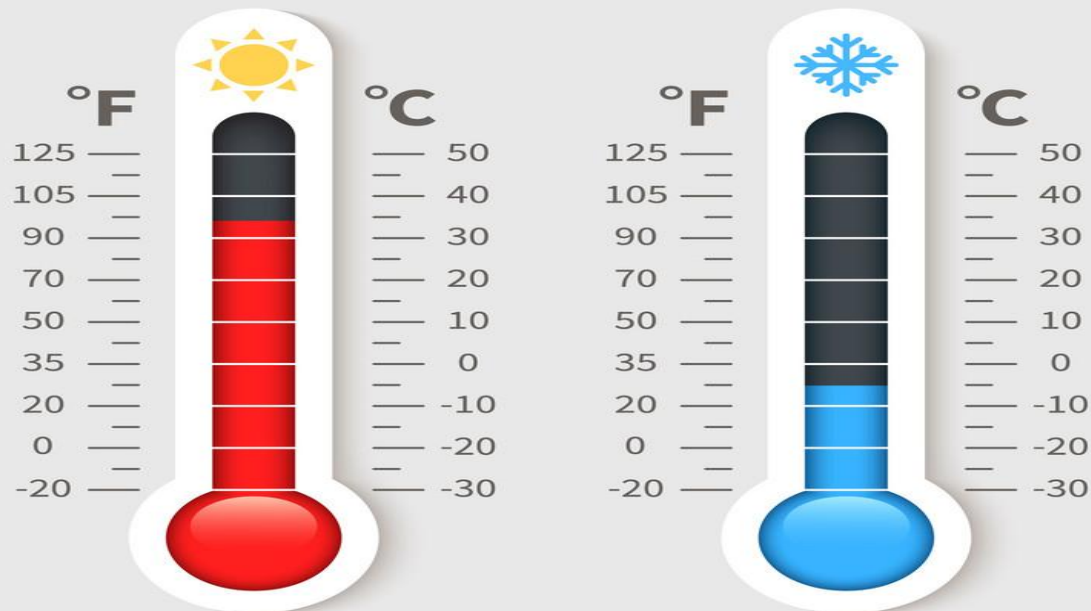


Definition:

Heat in a substance is called Temperature

OR

Temperature is the degree of sensible heat or cold, expressed in terms of specific scale.

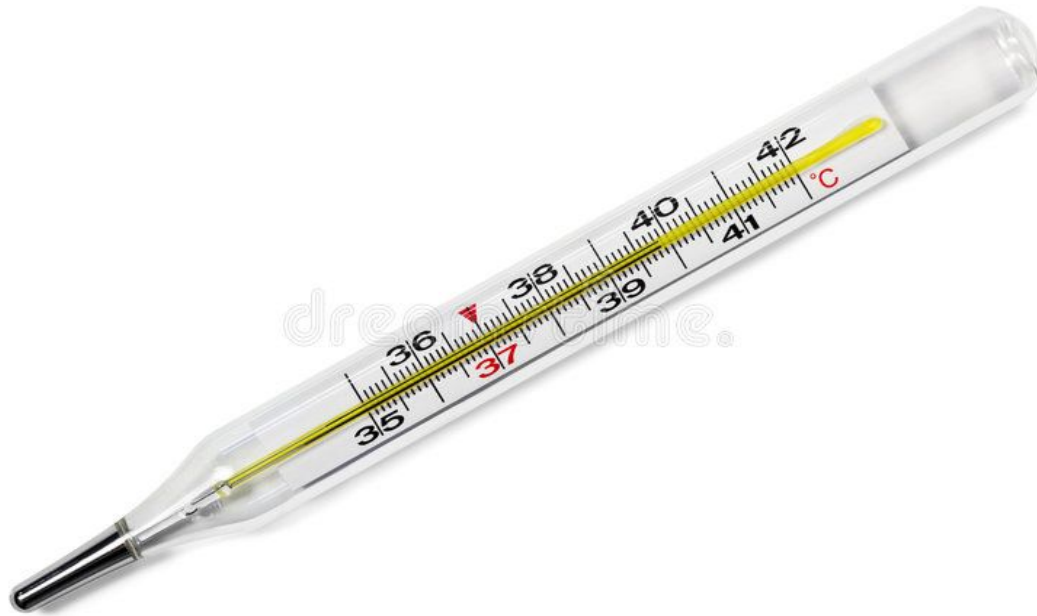


- It is a marker of endocrine, metabolic or muscle activity, the response of body to heat or cold in the environment; or the presence of infection or inflammation, among other illnesses. It is one of the vital signs.
- Body temperature is usually measured by thermometer placed in the mouth, the rectum or the auditory canal.
- Normal oral temperature is 37°C, rectal is 37.3°C

THE MOST COMMON SITES OF MEASURING BODY TEMPERATURE



TYPES OF THERMOMETERS





*Pasted on the forehead.
Corresponding temperature bar (black)
It will change colors.
Simple and convenient*



WHY DO WE CHECK TEMPERATURE?

- To detect abnormally low temperature.
 - To detect abnormally high temperature.
 - To detect fever.
 - To monitor the effectiveness of a fever-reducing medicine.
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BODY TEMPERATURE

Body Temperature:-

It is the difference between the amount of heat produced by the body and the amount of heat lost to the environment measured in degrees.



NORMAL BODY TEMPERATURE

- **Normal body temperature:** For healthy young adults the average body temperature is **37° C (98.6° F)**. The body tissues and cells function best in the normal body temperature. The normal range of the body temperature is between **36.5 to 37.5 C°**. **36.0C (97.0F) to 37.5C (99.5F)**.
- **Body temperature record:** Body temperature is recorded either in degree centigrade (C) or degree Fahrenheit (F).

- To convert Centigrade to Fahrenheit,

$$F = (9/5 \times C) + 32$$

$$\text{Ex. } 9/5 \times 37 + 32 = 98.6 \text{ F}$$

- To change Fahrenheit to Centigrade

$$C = (F - 32) \times 5/9$$

$$\text{Ex. } 98.6 - 32 \times 5/9 = 37 \text{ C}$$

PHYSIOLOGY OF BODY TEMPERATURE

- The core body temperature of a healthy person is maintained within a fairly constant range by the thermoregulatory set point of the **the thermo regulatory center in the Hypothalamus**.
- This center receives messages from cold and warm thermal receptors located throughout the body, compares that information with its temperature set point, and initiates responses to either produce or conserve body heat or to increase heat loss

HEAT PRODUCTION

The primary source of heat in the body is metabolism, with heat produced as a by product of metabolic activities that generate energy for cellular functions. Various mechanisms increase body metabolism, including hormones and exercise.

When additional heat is required to maintain balance, epinephrine and norepinephrine (sympathetic neurotransmitters) are released to rapidly alter metabolism so that energy production decreases and heat production increases.

HEAT PRODUCTION

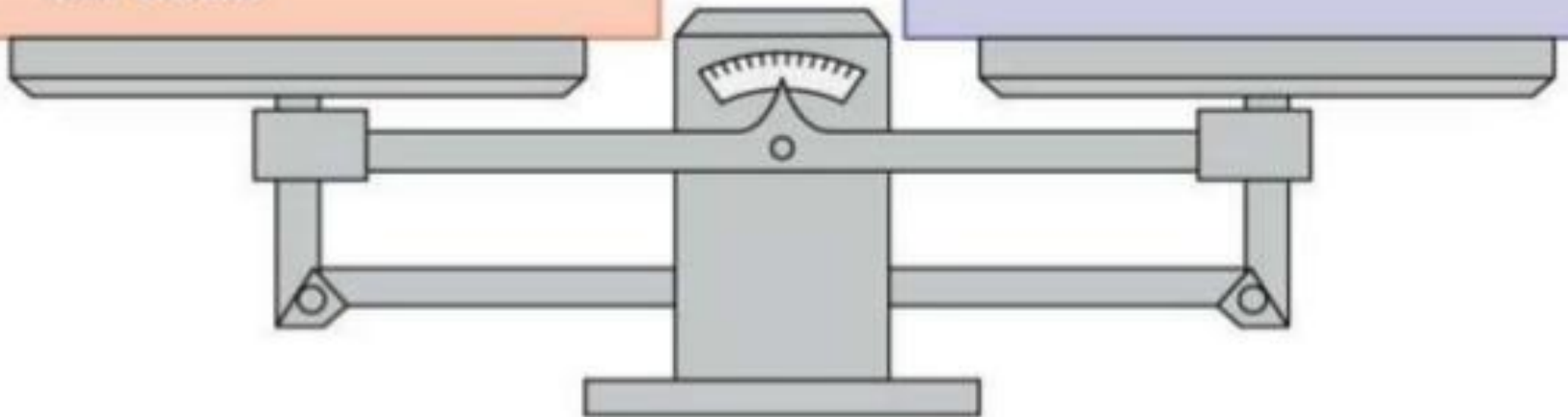
- **Thyroid hormone, produced by the thyroid gland, also increases metabolism and heat production, but over a much longer time period.**
 - **Shivering, a response that increases the production of heat, is initiated by the hypothalamus and results in muscle tremors.**
 - **Exercise increases heat production through muscle movements.**
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Heat production

- Basal metabolism
- Muscular activity (shivering)
- Thyroxine and epinephrine (stimulating effects on metabolic rate)
- Temperature effect on cells

Heat loss

- Radiation
- Conduction/convection
- Evaporation



FACTORS AFFECTING BODY TEMPERATURE



AGE

HORMONES

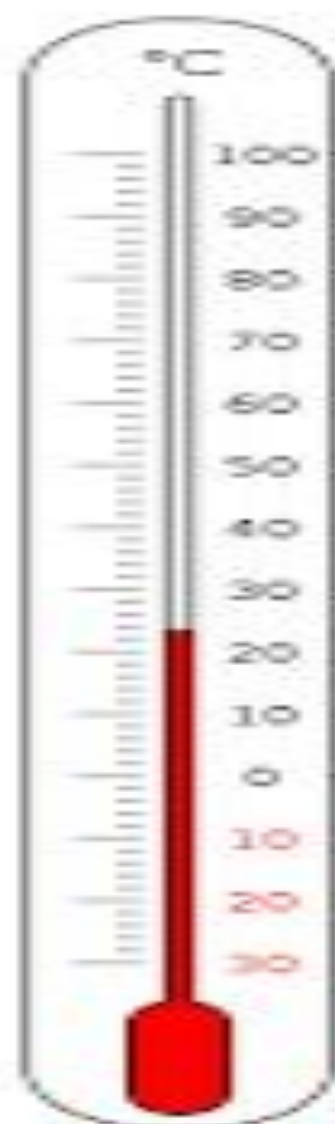


STRESS

ENVIRONMENT



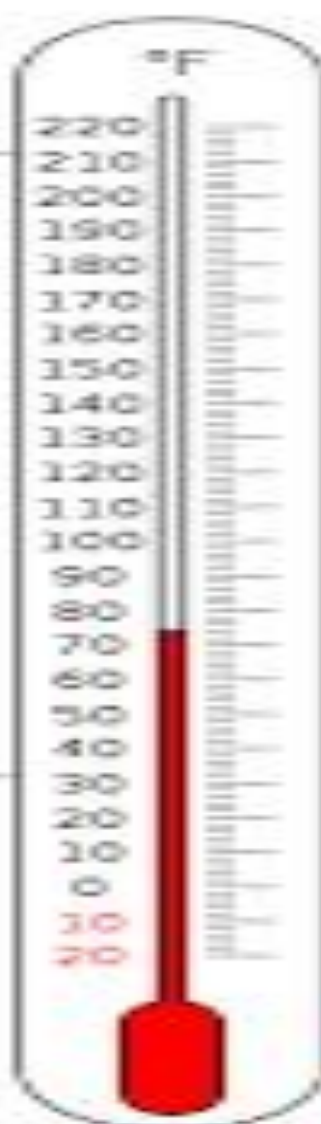
INFECTION



CELSIUS

BOILING POINT
100°C -- 212°F
OF WATER

FREEZING POINT
0°C -- 32°F
OF WATER



FAHRENHEIT

TEMPERATURE MEASUREMENT

Oral Temperature

Oral temperature is measured by placing the clinical thermometer under the tongue with mouth closed for a period of one minute.

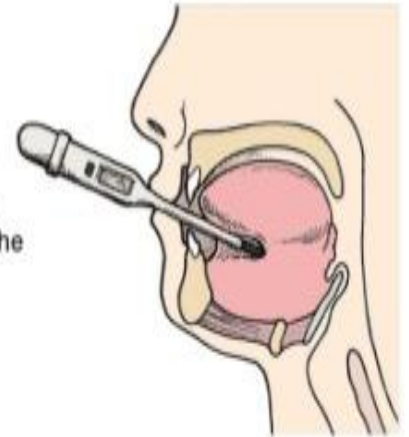
Prior to measurement no oral feeds for 15 minutes, wash the thermometer with antiseptic solution, mercury in the thermometer is gently shaken down.

How to Measure Body Temperature: Oral

1. Turn on thermometer according to package directions.



2. Place the tip of the thermometer under one side of tongue toward the back. Close mouth and breathe through nose.



3. Remove the thermometer after you hear the signal (usually a series of beeps) and read the temperature on the screen.



A fever is a temperature over 99.5 °F.

Axillary Temperature

The temperature obtained by placing a thermometer in the apex of the axilla with arm pressed closely to the side of the body for time recorded by the manufacturer of the thermometer.

The temperature obtained by this method is usually 0.5° to 1.5°F lower than oral.

How to Measure Body Temperature: Axillary

1. Turn on thermometer according to package directions.



2. Place the thermometer in a dry axilla.



3. Close the axilla by holding the elbow against the chest.



4. Remove the thermometer after you hear the signal (usually a series of beeps) and read the temperature on the screen.
A fever is an axillary temperature over 99 °F (37.2 °C).

Rectal Temperature

The temperature obtained by inserting a thermometer into the anal canal to a depth of at least 3.8 cm and holding it in place for 3 to 5 minutes.

A rectal temperature is more accurate than either oral or axillary temperatures.

It averages about 1°F higher than the oral temperature and approx. 1.5°F higher than the axillary temperature.



A rectal temperature can be taken in either of these positions.

Tympanic Temperature

The temperature obtained by placing an electronic probe in the ear canal. Such a reading measures the temperature in the capillary bed of the tympanic membrane and is generally reflective of the core temperature.

ALSO CALLED PYREXIA

**Is a high temperature
above normal range :**

$> 37.3^{\circ}\text{C}$

HYPERPYREXIA

**Is a very high
temperature above :**

41°C

HYPOTHERMIA

**Is low temperature
below normal range :**



Temperature Basics

°F °C

100.4 38

98.6 37

98.8 36

95.0 35

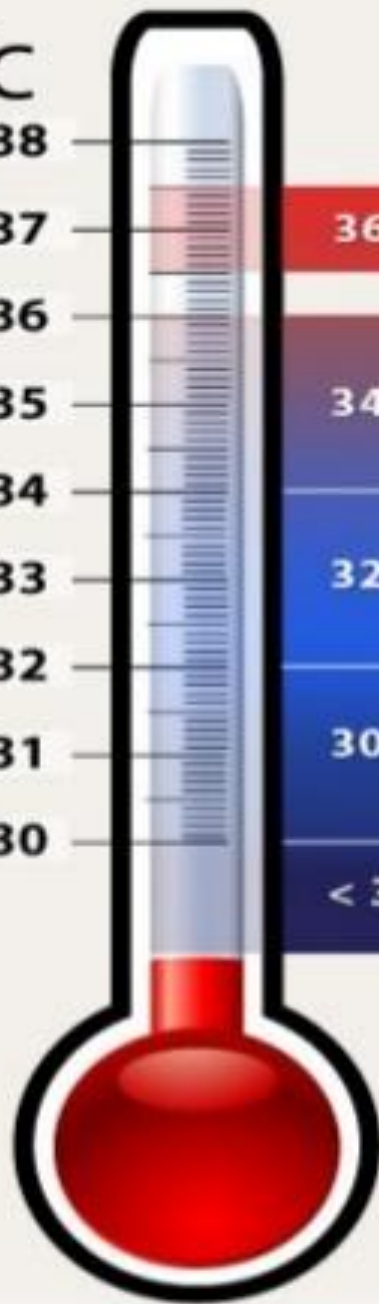
93.2 34

91.4 33

89.6 32

87.8 31

86.0 30



36.5-37.5°C Normal Adult Body Temperature

34-36°C Mild Hypothermia

32-34°C Moderate Hypothermia

30-32°C Moderate/Deep Hypothermia

< 30°C Deep Hypothermia